

What are the inputs?

- 1: What use cases are covered?
- 2: Do we have representation of difficult inputs in the tail?
- 3: Are the inputs adapted to the model (e.g. multi-turn)?

How do you call the language model?

- 1: How do you prompt the language model?
- 2: Does the language model use chain-of-thought, tools, RAG, etc?
- 3: Are we ~~evaluating~~ evaluating the language model ~~model~~ or an agentic system (model developer wants former, user wants latter)?

How do you evaluate the outputs?

- 1: Are the reference outputs used for evaluation error-free?
- 2: What metrics do you use (e.g. pass@K)?
- 3: How do you factor in cost (e.g. inference + training)?
- 4: How do you factor in asymmetric errors (e.g. hallucination in a medical setting)?
- 5: How do you handle open-ended generation (no ground truth)?

How do you interpret the metrics?

- 1: How do you interpret a number (e.g. 91%) - is it ready for deployment?
- 2: How do we assess generalization in the face of train-test overlap?
- 3: Are we evaluating the final model or the method?

def perplexity(C): $\frac{1}{|C|}$

Recall: A language model is a probability distribution $p(x)$ over a sequences of tokens.

Perplexity: $(1/p(x))^{1/|D|}$ measures whether p assigns high probability to some dataset D .

⌋ In pre-training: you minimize perplexity on the training set.

~ The obvious thing is to measure perplexity on the test set.

~ $\Rightarrow$ Standard dataset: WikiText-103 (wikipedia), Penn Treebank (WSJ)
decontamination: train-test overlap.

~ $\Rightarrow$ GPT 2: trained on webText (40GB text, linked from reddit), zero-shot on standard datasets.

~ $\Rightarrow$ GPT 2 / GPT 3: shifted towards more downstream task accuracy, but perplexity is still useful:

①: Smoother than downstream task accuracy (fitting scaling law)

②: is universal (why we use it for training) whereas task accuracy might miss some nuances

③: note: can measure conditional perplexity on downstream task too (used for scaling law)

Caveats: if you're running a leaderboard: evaluator needs to trust the language model.

⌋

2. how do we ...